

Subiksha Jegadish

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EDUCATION

University of Maryland, College Park

College Park, MD

Master of Engineering in Software Engineering | GPA: 4.0/4.0

Aug 2025 – May 2027 (Expected)

- Coursework: Software Engineering; Software Testing & Maintenance; AI-Based Software Systems; Fundamentals of AI and Deep Learning

Vellore Institute of Technology

Vellore, India

Bachelor of Technology in Computer Science and Engineering | CGPA: 9.06/10.0

Aug 2020 – May 2024

EXPERIENCE

Software Development Engineer Intern

Arlington, VA

Amazon Web Services

Jun 2026 – Aug 2026

- Shipped an S3-triggered data validation layer (Python, PySpark on **AWS Glue**) validating **2–3M+ rows in under 10 minutes** across 5+ checks, replacing 24-hour-late failures with immediate vendor feedback
- Owned infrastructure end-to-end in **AWS CDK** – Glue orchestration, CloudWatch alarms/dashboards, and a ticket-cutting Lambda – onboarding **15+ internal AWS teams** (EC2, S3, and others) across Beta, Gamma and Production.
- Protected recommendation accuracy for **99.8%** of downstream pipeline runs by blocking malformed vendor drops upstream, and cut invalid-file rejections by **97%** after redesigning rejection logic

Software Analyst

Bengaluru, India

ION Group (Global Fintech – Trading & Risk Management)

Jan 2024 – Jul 2025

- Engineered standardized **CI/CD pipelines** using **Jenkins + Docker** for trading platforms used by **50+ financial institutions**, reducing release cycles from **5 days → 2 days**
- Reduced deployment failures by **78%** through pipeline hardening, automated checks, and environment standardization
- Resolved **100+ production incidents** via **SQL performance tuning**, diagnostics, and RCA, improving MTTR by **35%**
- Automated database backup and restore workflows using **Python + Bash**, cutting environment setup from **72 hrs → 3 hrs** across **50+** client environments

Software Development Intern

Bengaluru, India

ISRO – Liquid Propulsion Systems Centre

May 2023 – Jul 2023

- Developed an end-to-end **computer vision pipeline** (Python, OpenCV, scikit-learn) for welding defect detection, achieving **93.8% accuracy** on **500+** samples
- Implemented ML-based anomaly detection (Random Forest, SVM) to identify **150+ critical defects**, improving quality-control reliability by **38%**

PROJECTS

Distributed Event Management System | React, Node.js, Express, PostgreSQL, Docker

Sep – Dec 2025

- Architected a scalable **microservices-based event platform** with **JWT authentication**, **RBAC**, and async notification workflows supporting **1000 +** concurrent multi-role users
- Built RESTful APIs with **Node.js + Express**, **PostgreSQL** connection pooling, **Docker**, and real-time QR check-in via **Socket.io**, cutting check-in time to **90 sec**

Algorithmic Trading Strategy Simulator | Python, FastAPI, React, Docker

Nov – Dec 2025

- Built a full-stack backtesting platform computing **CAGR**, **Sharpe Ratio**, and **Max Drawdown** across **10** strategies, exposed via **FastAPI** and an interactive **React** dashboard
- Containerized with **Docker**, cutting backtest runtime by **60%** | **GitHub:** github.com/subikshajegadish/algo-trading-sim

Differentiable NAS for Optimizing Vision Encoders in Multimodal Models | PyTorch, DARTS/GDAS2025

- Implemented a DARTS-style NAS framework to optimize ViT-style vision encoders over depth, attention heads, MLP ratios, patch sizes, and skip connections
- Integrated multi-objective efficiency penalties (FLOPs, parameter count, visual token count) and validated on a filtered **MSCOCO VQA_{v2}** subset (**309 Q&A** across **50** images)
- Real-data search favored shallower architectures (**6** layers vs **8**) with more skip connections, achieving **13.8M** params and **6.8** GFLOPs

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, JavaScript, C/C++, SQL

Backend & Web: FastAPI, Django, Node.js, Express, React

Data & ML: Pandas, NumPy, scikit-learn, OpenCV

DevOps & Tools: Git, Docker, Jenkins, AWS (Lambda, Glue, Step Functions, EventBridge, EC2, RDS), Linux, Postman, pytest

Concepts: REST APIs, Microservices, CI/CD, TDD, System Design, Observability, Agile